

Tory Seaton
FA 550 Capstone Project

Deliverable 2:

Section 1: Executive Summary

Project Title: The Hidden Crisis: How Substance Use and Mental Health Collide in America

Project Track: Data Storytelling / Narrative

Project Description: This project is an interactive data dashboard that combines eight connected stories about the substance abuse crisis in the United States. The project references four government datasets from the Substance Abuse and Mental Health Services Administration, totaling over 4.4 million individual records. Specifically, the dashboard explores how substance use and mental health intersect across multiple demographics such as age, gender, and race. It also tracks the effectiveness of different treatment methods for substance abuse recovery, such as pre-treatment employment rates compared to post-treatment employment rates. I chose substance abuse and recovery as my dashboard topic because it is a persisting issue that many communities worldwide are facing and is a growing issue that needs to be addressed.

Key Features:

- Eight interactive chapters covering the full arc of the crisis, from the scale of substance use in America to treatment outcomes and the barriers that prevent people from receiving help
- Year filter buttons on every chapter allowing users to explore trends across 2021–2024
- Hover tooltips on every chart providing exact data points
- Animated stat cards, line toggle buttons, and before-and-after comparisons that make complex public health data accessible to a non-technical audience
- A self-reporting caveat and data limitation notes built directly into the dashboard so users understand the boundaries of what the data can and cannot tell us

Technologies used:

- Python — pandas and pyreadstat libraries for data cleaning, decoding, and processing
- HTML / CSS / JavaScript — full front end dashboard built from scratch
- Chart.js — all eight visualizations rendered using Chart.js v4.4.1
- Claude AI (Anthropic) — used throughout for data exploration, code generation, debugging, and storytelling
- SAMHSA Public Data Portal — source for all four datasets

Key Findings:

1. Americans aged 21–25 represent the highest risk window for substance abuse — alcohol peaks at 58.3%, nicotine vaping at 26.7%, and depression at 17.1% at the exact same life stage. This suggests a generation navigating major transitions without adequate mental health support.
2. White Americans receive mental health treatment at nearly double the rate of Black Americans (32.0% vs 17.9%). This can be tied to a systematic issue where underrepresented groups do not have the same medical access.
3. Less than 40% of people who enter substance abuse treatment successfully complete it. The main driver for this is the cost of treatment, followed by the lack of awareness on where to go for treatment.

Impact/Value:

This project benefits two audiences: public health officials and the general public. For public health officials, this dashboard serves as clear, visualized evidence providing a holistic view of substance abuse in America — covering the physical and mental effects while also providing information on the recovery process and where fixes can be made to better the experience of individuals seeking help. For the general public, this dashboard can be used to raise awareness of this crisis by providing one platform where people can access the information they need to learn about substance abuse in their communities. Many people have been affected by substance abuse whether they have personally experienced it or watched a loved one go through it.

Section 2: Project Goals & Context**Problem Statement:**

Substance abuse is one of the most prevalent public health crises in the United States, directly affecting more than 50 million Americans. It is a leading cause of preventable deaths and illness, carries extreme negative effects on mental health and the economy, and impacts workforce productivity and the overall functioning of communities. Its effects on youth are also rising at an alarming rate. Substance abuse does not affect just one type of person — it works like a domino effect, spreading through families and communities in ways that are hard to contain but possible to address. When individuals get the help they need, entire communities can grow and thrive. Because substance abuse can happen to anyone regardless of age, background, or income level, it is an issue that deserves accessible information provided by resources such as this narrative dashboard.

Problem: Objectives & Target Audience:

The goal of this project is to build an interactive dashboard that tells the bigger picture of substance abuse in America. I want to steer people away from only having stigmatized views and little information on substance abuse as a whole. It is a real issue that can impact anyone's lives at any moment in their life. This dashboard is designed to be informative and to highlight specific issues that public health officials can work to address, such as the high frequency of substance use among younger age groups and the lack of affordable and accessible treatment options. For the general public, this dashboard is meant to show what is really happening and make it clear that this issue is everywhere and affects families, communities, mental health, the economy, and the workforce.

Most people are not aware of the treatment and recovery process unless they have personally experienced it firsthand or watched someone close to them go through it. One of the goals of this project for the general public audience is to provide them with a different perspective that uses real data and not just popular assumptions. My dashboard is presented in a way that is both professional and formal for public health officials to review and take serious as an information source, but also easy to understand for the general public. The end goal is that after going through this dashboard both audiences leave with a clearer understanding of the crisis and a stronger sense of why action is not an option.

Personal Motivation:

I chose substance abuse as my topic because it's a topic I have been interested in since I was younger. Growing up I watched a lot of VICE drug documentaries that followed the entire process of the drug trade worldwide, but would only briefly touch on the more serious aspects of the people affected such as treatment success rates, mental health effects, and social consequences. That always left me curious about how widespread this issue truly is and which methods are genuinely effective in the recovery process. I am also drawn to topics where underrepresented groups receive less support from the systems designed to help them, and substance abuse is a clear example of that. Particularly, when it comes to access to treatment across different racial and economic groups in America. That combination of personal curiosity and a desire to shed light on an underreported side of this crisis is what made this the right topic for my capstone project.

Section 3: Data Preparation Process

Data sources:

All four datasets used in this project were collected and published by the Substance Abuse and Mental Health Services Administration (SAMHSA), which is a division of the U.S. Department of Health and Human Services. All datasets are free and publicly available at samhsa.gov/data.

The primary dataset is the National Survey on Drug Use and Health (NSDUH) with data collected from 2021 to 2024. This is a nationally representative household survey where Americans aged 12 and older are asked directly about their substance use, mental health, and treatment experiences. It contains 232,441 individual survey respondents, 2,638 variables, and covers the years 2021 through 2024. The file was downloaded in Stata (.dta) format.

The three supplementary datasets are from the Treatment Episode Data Set Discharges (TEDS-D) covering 2021, 2022, and 2023 separately. TEDS tracks every admission and discharge from publicly funded substance abuse treatment facilities across the country. When combined, the three years contain 4,219,911 treatment episodes across 76 variables each. All three files were downloaded in Stata (.dta) format.

Data Quality Assessment

Overall, the data quality was relatively strong given that all four datasets came from an established federal government organization with rigorous data collection standards. The most significant issues found was that variables were not consistent across all four years of the NSDUH dataset. For example, mental health treatment data was only collected in 2022 and 2023, which is part of the reason I sought out additional datasets to provide more extensive information on the recovery side of the story. Nicotine vaping and cigarette data was also absent for the year 2021, which was surprising to me personally. As a college student in a city with a younger population, vaping is something I see everywhere, so finding out it was not tracked until 2022 highlighted just how recently it became a recognized public health priority worth looking deeper into.

Another major challenge was that every variable in the original NSDUH dataset is stored as a numeric code. There are 2,638 columns and each one has its own coding system, meaning race, age, substance use, and every other variable is represented as a number with no label attached. Without decoding, the data was completely unreadable and impossible to visualize accurately.

Cleaning and transformation steps:

All data cleaning was done in Python using the pandas and pyreadstat libraries.

Step 1 — Loading the data: The NSDUH Stata file was loaded using pyreadstat with only the key columns needed for analysis rather than all 2,638. The three TEDS files were loaded the same way, pulling only 21 key columns per file to avoid exceeding memory limits. Loading all three TEDS years at full column width caused a memory crash during an earlier attempt.

Step 2 — Decoding numeric labels: All numeric codes were mapped to human readable labels using value label dictionaries extracted directly from the Stata file metadata. Age group 5 became "21–23", race 2 became "Black", alcohol use 1 became "Past 30 days" and so on across every key variable.

Step 3 — Handling special codes: Values of -9, 97, 98, and 99 were identified and excluded from all calculations. Fentanyl values of -9 in 2024 and vaping values of -9 in 2021 were replaced with null so they display as N/A in the dashboard rather than breaking the charts.

Step 4 — Year filtering: The NSDUH dataset was split into two versions — one containing all four years for substance use analysis and one filtered to 2022–2023 only for treatment and mental health analysis where those variables are available.

Step 5 — Creating binary flags: New binary columns were created for every key metric — `uses_alcohol`, `uses_marijuana`, `uses_cocaine`, `uses_heroin`, `uses_fentanyl`, `uses_cigarettes`, `uses_vaping`, `has_depression`, `got_mh_treatment` — coded as 1 or 0 to enable clean percentage calculations across demographic groups.

Step 6 — Pre-aggregation: All chart values were pre-calculated in Python by year, age group, gender, and race and hardcoded directly into the dashboard JavaScript. This eliminated any need for client-side data processing and ensured the dashboard loads instantly with no performance issues.

Tools used: Python, pandas, pyreadstat, JavaScript, Chart.js

Section 4: Design Decisions & Rationale

Visualization	Type Chosen	Alternatives Considered	Why This Type	Interactivity
Visualization 1: The Scale of the Crisis	Horizontal bar chart with animated stat cards	Pie chart, vertical bar chart	The horizontal bars make substance labels easy to read. The stat cards allows user to see the four most important takeaways from the page.	Year filter buttons (All Years, 2021, 2022, 2023, 2024) Hover tooltips on every bar Animated stat cards that count up from zero on page load
Visualization 2: Young Adults Are the Eye of the Storm	Multi-line chart	Grouped bar chart, area chart	The line-chart show how multiple substances rise and fall across 11 age groups simultaneously. This allows users to immediately track correlated trends and spot peaks for specific age groups.	Year filter buttons (All Years, 2021, 2022, 2023, 2024) Five individual line toggle buttons for alcohol, marijuana, cigarettes, nicotine vaping, and depression Hover tooltips on every data point
Visualization 3: Men Use More. Women Ask for Help More.	Side by side animated bar chart	Grouped bar chart, radar chart	The two column layout places male and female directly next to each other making differences visible at first glance.	Year filter buttons (2022 and 2023) All bars animate in from zero on first page visit
Visualization 4: Same Crisis. Different Access.	Three separate vertical bar charts	Single grouped bar chart, heat map	Combining all three into one chart would make the visulization too cluttered to read. Separate charts let each finding land clearly.	Year filter buttons (2022 and 2023) Hover tooltips showing exact rates per group Footnote flagging Native American sample size
Visualization 5: Does Treatment Work?	Donut chart paired with horizontal bar chart	Pie chart, stacked bar, table	The donut chart shows overall proportions clearly. By adding the horizontal bar chart, completion ranks based on substance	Year filter buttons (2021, 2022, and 2023) that update both charts simultaneously Hover tooltips

			are immediately recognized by users.	showing exact completion and dropout rates per substance
Visualization 6: Not All Treatment Is Created Equal.	Horizontal bar chart	Vertical bar chart, table	The eight treatment types have long names. A horizontal layout makes the labels readable.	Year filter buttons (2021, 2022, and 2023) Hover tooltips on every bar showing the completion rate and description of each treatment type
Visualization 7: Treatment Works — When People Can Access It.	Two side by side before and after bar charts	Single grouped bar chart, line chart	The side by side charts makes it easier for users to view the improvements in employment and arrest rates.	Year filter buttons (2021, 2022, and 2023) Hover tooltips showing exact rates at admission and discharge
Visualization 8: The Treatment Gap — Who Gets Left Behind.	Donut chart paired with bold text callout	Bar chart, word cloud, infographic	The donut chart shows overall proportions clearly. The text callout delivers the policy message directly without adding more data.	Hover tooltips on every donut slice showing exact percentages Four animated stat cards showing total people who needed treatment but did not receive it and the top three barriers

Encoding strategies :

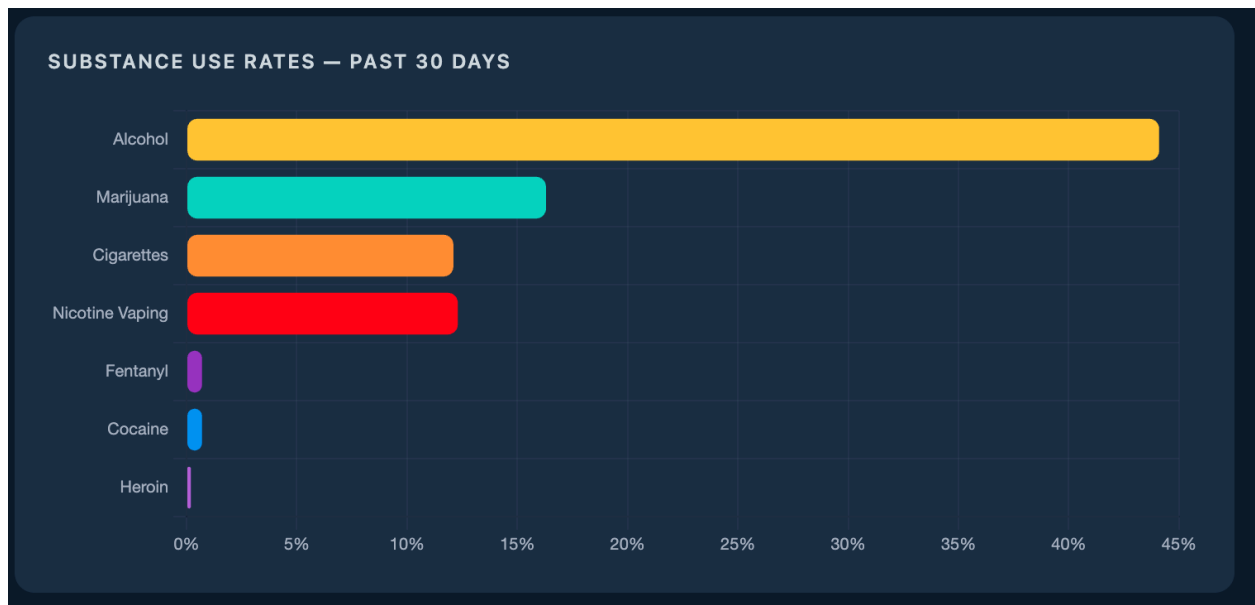
The dashboard uses a consistent color system for each substance throughout the entire dashboard. All visualizations with substance as a variable contain the same color so users can recognize which substance is which as they move through the story. Other visualizations such as Overall Treatment Outcomes and Substance Use by Race use bright contrasting colors that are completely different from the bar or color representation next to it, allowing users to easily differentiate what each section is showing. For the stat cards, gold is used for primary statistics and highlights, red is used for negative or alarming findings, and teal is for positive outcomes such as treatment completion. The scales of each chart are dependent on the number of visualizations each page has, but all are easy to view and proportionate to the overall look of the dashboard.

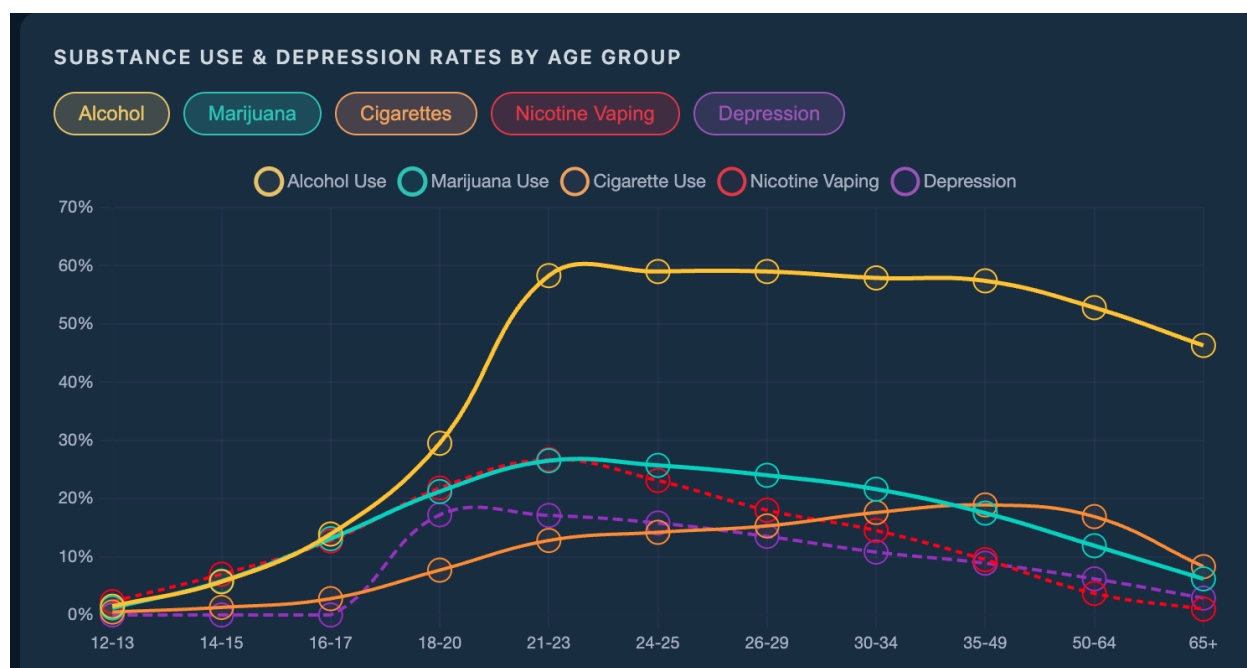
Layout and organization:

I wanted the dashboard to reveal small pieces of a bigger picture with each page. I started with broader visualizations such as overall substance use rates and gradually moved into more specific stories such as data broken down by race and age. Users can navigate between pages by clicking the gold button labeled with an arrow at the bottom of each page, using the arrow keys on their keyboard, or by clicking the progress dots at the top of the dashboard.

Screenshots:

Used matching colors for all matching substances to create a pattern recognition for users when they see the same substances represented across different charts*





Section 5: Technical Implementation

Technology stack:

I used Python 3.12 for all data cleaning and processing. I chose Python because it is the most common coding language for working with large datasets and has libraries specifically designed for reading Stata files. The two key libraries were pandas 2.2.0 for data manipulation and pyreadstat 1.2.6 for reading the Stata .dta files. I considered using R for the data cleaning but stuck with Python because it was I am more familiar with.

HTML, CSS, and JavaScript were used to build the actual dashboard. I chose to build it in web technologies rather than using Tableau or Power BI because it gave me complete control over the design and interactivity aspects of the dashboard.

Chart.js 4.4.1 was used for all eight visualizations. I chose Chart.js because it handles and creates animated charts cleanly and at a quick pace.

Claude AI (Anthropic) was used throughout the entire project for data exploration, cleaning script generation, code writing, debugging. It is fast and responsive and also assists with providing thorough feedback on what changes can be made.

GitHub Pages was used to deploy and host the final dashboard as a live website. I chose GitHub Pages because it is free and compatible with static HTML files.

Structure:

The project is organized as a single HTML file (index.html) that contains all the HTML structure, CSS styling, and JavaScript logic in one place. This kept the project simple and portable. Anyone can download one file and open it in a browser and the entire dashboard works with no necessary setup.

Inside the file, the HTML section defines nine pages (a landing page and eight chapter pages). The CSS section styles everything using CSS variables for colors so the entire color scheme can be changed by editing a few lines. The JavaScript section handles the navigation between pages, the data for every chart pre-calculated from Python, and the Chart.js chart building functions that run when each page is visited.

The data flow starts with the raw SAMHSA datasets that were cleaned in Python and all the numbers for every chart are pre-calculated and organized by year and demographic group. Those numbers are then hardcoded directly into the JavaScript as data objects. When a user navigates to a chapter the JavaScript reads the relevant data object and builds the chart. When a user clicks a year filter button the JavaScript updates the chart with the new data for that year. Nothing is loaded from an external source at runtime which is why the dashboard loads instantly.

Key Technical Implementations:

The most interesting technical solution was the year filter system. Each chapter has its own active year stored in a JavaScript object called activeYear. When a user clicks a year filter button it updates that chapter's active year and calls a function called buildOrUpdate which either builds the chart for the first time or updates the existing chart with new data. This meant I did not have to rebuild charts from scratch every time a filter was clicked — Chart.js updates the existing chart with a smooth animation.

Memory management was a challenge that required a specific solution. Loading all three TEDS years at full width caused a 4GB memory crash. The fix was to load only 21 key columns per file instead of all 76, combine the three years, then delete the individual year dataframes from memory immediately after combining. This reduced peak memory usage enough to process 4.2 million rows without crashing.

Code Samples:

Sample 1 : Year filter update function (JavaScript) This function runs every time a user clicks a year filter button. It updates the active year for that chapter and rebuilds the chart with the new data.

```
function setYear(ch, yr, btn) {  
  // Update the active year for this chapter  
  activeYear[ch] = yr;  
  
  // Remove 'active' class from all buttons in this filter bar  
  // and add it to the one that was just clicked  
  btn.closest('.year-filter-bar')  
    .querySelectorAll('.yr-btn')  
    .forEach(b => b.classList.remove('active'));  
  btn.classList.add('active');  
  
  // Rebuild or update the chart with the new year's data  
  buildOrUpdate(ch);  
}
```

Sample 2 : Decoding numeric labels (Python) This runs during data cleaning and converts every numeric code in the NSDUH dataset into a human readable label. Without this step the data is completely unusable.

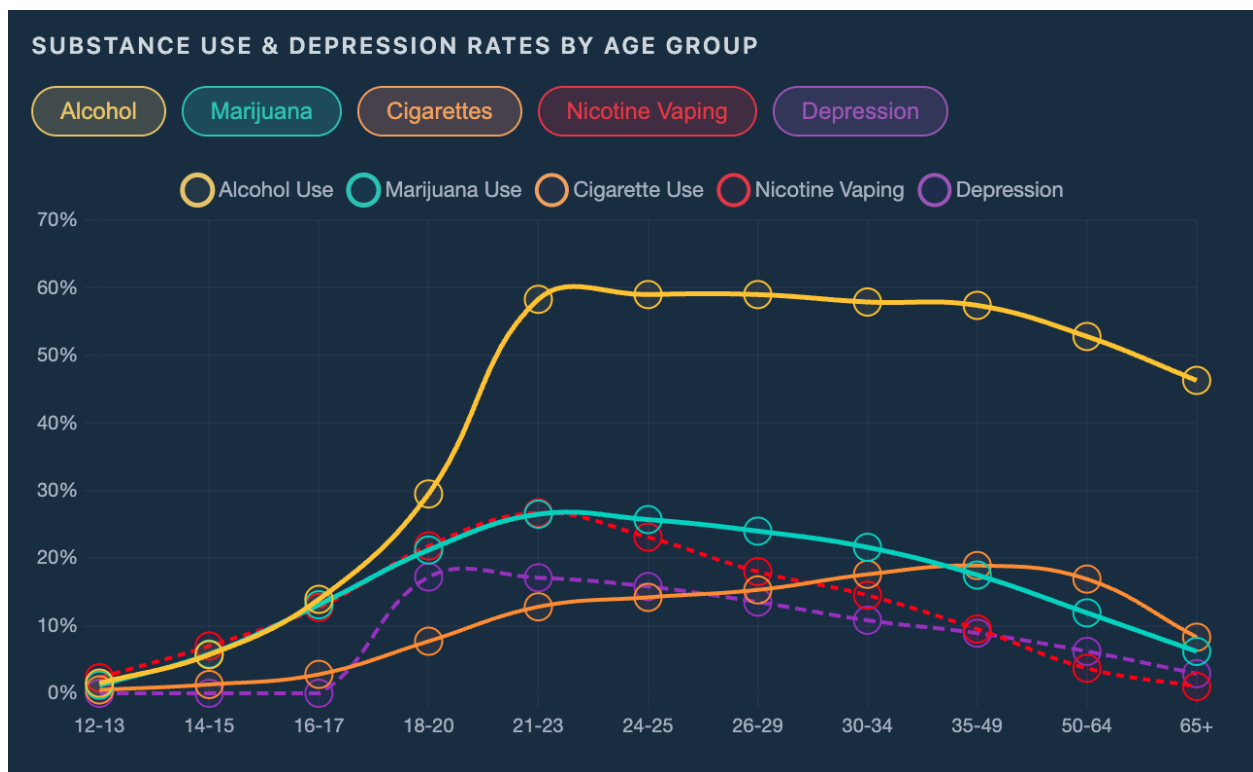
```
# Mapping dictionaries — each number maps to its real meaning  
clean_race = {  
  1: 'White',    2: 'Black',  
  3: 'Native American', 4: 'Pacific Islander',  
  5: 'Asian',    6: 'Multiracial',  
  7: 'Hispanic'  
}  
  
clean_age = {  
  1: '12-13', 2: '14-15', 3: '16-17',  
  4: '18-20', 5: '21-23', 6: '24-25',  
  7: '26-29', 8: '30-34', 9: '35-49',  
  10: '50-64', 11: '65+'  
}
```

```
# Apply the mappings to create new readable columns
df['race'] = df['NEW_RACE2'].map(clean_race)
df['age_group'] = df['AGE3'].map(clean_age)
```

Section 6: Key Findings & Insights

Major findings:

Finding 1:



Americans between the ages of 21 and 25 are the highest risk group for substance abuse. Based on my findings, when depression rates peak at 17.1%, alcohol peaks at 58.3%, nicotine vaping at 26.7%, and marijuana at 26.5% all at the exact same life stage. This correlation is extremely important because it can be used to inform preventative measures and ensure that young audiences receive education on substance abuse before these patterns continue to grow. The overlap between substance use and depression in this age group suggests that many young adults may be turning to substances to cope with mental health issues rather than seeking professional help, which points to a gap in both mental health awareness and accessibility for this age group.

Finding 2:



32% of White Americans received mental health treatment compared to 17.9% of Black Americans, despite both groups having similar substance use rates. The disparity in treatment access is not explained by differences in use, which makes it a systemic issue rather than a personal one. Other underrepresented communities including Hispanic and Asian Americans also show treatment rates below the national average. This finding is meant to bring attention to the fact that some groups have significantly more access to and awareness of treatment options than others. The goal of including this visualization is to shed light on this inequality and hopefully contribute to conversations about how equal opportunity and access to care can be extended to all groups.

Finding 3:

The Treatment Gap — Who Gets Left Behind.

Even when people recognize they need help, millions never receive it. The barriers keeping people from treatment are not personal failures — they are systemic failures that policymakers have the power to fix.

1,003

Respondents who needed treatment but didn't receive it

33%

Said cost was the reason they didn't seek help

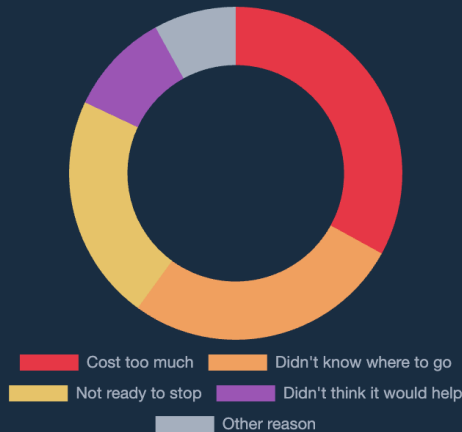
27%

Didn't know where to go for treatment

22%

Were not ready to stop using

WHY PEOPLE DIDN'T SEEK TREATMENT



Cost is the #1 barrier to treatment.

One in three people who needed treatment and didn't get it cited cost as the reason. This is not a personal decision — it is a policy failure. Expanding insurance coverage, funding community treatment centers, and removing financial barriers would directly reduce this number.

"The data is clear. The solutions exist. The only thing missing is the political will to act."

Based on 1,003 respondents from the NSDUH dataset who needed treatment but did not receive it, 33% cited cost as the primary reason, making it the most common barrier to seeking help. The second most common reason was not knowing where to go for treatment at 27%. This finding is important because it challenges the assumption that people who are not receiving help are simply choosing not to get better. In reality many people want help but are blocked by a system that lacks awareness, affordability, and accessibility for people across all socioeconomic backgrounds. These are not personal failures. They are gaps in the system that policy decisions can directly address.

Patterns and trends:

A common pattern that not enough people are aware of is the general shift in nicotine and tobacco products. From 2022 to 2024 cigarette use peaks in the 35 to 49 age group at 18.9% while vaping peaks at 26.7% in the 21 to 25 age group. While it is essentially the same substance, different generations are still using it through a different device. The same way cigarettes were frequently used by older generations and took over their lives, a similar pattern is happening to younger generations just through a different medium. The acknowledgment of this pattern should be alarming to public health officials and action needs to be taken to prevent it from progressing.

Another trend regarding cigarette and vape usage is the overwhelmingly high use of these products among Native Americans. 21.5% of Native Americans reported smoking cigarettes and 18.4% reported using vapes, with the average usage across all other racial groups being closer to 13%. This is a pattern I was previously unaware of and I am curious about what factors play into that.

Another notable pattern found in my dashboard is that women seek mental health treatment at a 33.1% rate compared to men who seek treatment at a 19.9% rate. This may be due to the frequently supported idea that men should not need to take care of their mental health or that showing emotion is seen as weakness. I think it is important to note how social norms and double standards can have such a big impact on people, preventing them from getting treatment that could possibly change their lives for the better.

Implications:

The findings in this dashboard point to several systemic issues that can be addressed through policy changes. Access to treatment and resources is being limited by racial, economic, and geographic disparities across America. Specifically, restructuring targeted funding for community based treatment centers, expanding insurance coverage for substance abuse treatment, and ensuring that all communities, especially those with high rates of substance abuse, have access to the resources they need are all steps that could make a meaningful difference.

Public health officials should care because it is their obligation and responsibility to provide the necessary resources for the communities where help is most needed. They have the tangible power to make both internal and external changes and have a duty to serve all members of their communities equally. The general public should care because substance abuse affects

communities on every level and many people have personal connections to this issue whether they realize it or not.

Limitations:

A really important caveat that I thought was necessary to include is that the reported rates in this dashboard are most likely underestimates. Because substance use is a sensitive topic surrounded by stigma, social pressure, and legal consequences, some respondents may not have been truthful in their responses. The actual prevalence of substance use across all substances is likely higher than what is shown in the dashboard.

Another limitation is that the NSDUH does not include incarcerated or homeless populations, which makes the data less representative of the full scope of the issue than I would have preferred. These are some of the highest risk groups for substance abuse and their absence from the data means the crisis is likely even more widespread.

A limitation specific to the TEDS dataset is that it tracks episodes not individuals. If one person has sought out treatment three different times that is recorded as three separate entries, not as one person. This makes it difficult to track one individual's recovery journey over time or calculate true long term recovery rates.

Lastly, all findings in this dashboard can only be presented as correlations and not causations because the data was not sourced from randomized controlled experiments. The patterns are real and meaningful but the data alone cannot prove that one variable directly causes another.

Section 7: Challenges & Solutions

Challenge 1: Nicotine and Tobacco were missing from the dashboard

Problem: When reviewing my dashboard, I realized that nicotine and tobacco (two of the most widely used substances in America) were not included in the dashboard.

What I tried: Initially Claude just included the alcohol, marijuana, fentanyl, cocaine, heroin. I did not consider the possibility that it would leave out substances even though I prompted it to use all substances included in the dataset.

The solution: I instructed Claude to go back into the raw NSDUH dataset and scan all 2,638 column names for anything related to nicotine or tobacco. It found three key variables and once identified I added both substances to all four chapters of the dashboard.

How AI helped: Claude reviewed the dataset again and identified the right variables quickly. It also flagged that nicotine vaping data was not collected in 2021, which is why the vaping line shows as null when users filter to that year.

Challenge 2: Variables were not consistent across all four years

Problem: All four years of the NSDUH did not have the same variables. Mental health treatment data was only collected in 2022 and 2023, fentanyl data was missing from 2024, and nicotine vaping was not collected in 2021. When I ran analysis across all years without knowing this I was getting strange values that skewed my results.

What I tried: I initially loaded all four years together and ran calculations without checking if the variables matched. This provided results that were not readable.

The solution: I asked Claude to run a year by year audit of every key variable to identify exactly which years had complete data and to separate the datasets into two versions.

How AI helped: Claude helped me filter the data based on year to prevent the visualizations from missing any data. Instead, if data was not collected during a specific year the stat card says N/A and includes a footnote explaining why.

Challenge 3: Every variable was a number with no labels

Problem: Every variable in the original NSDUH dataset is stored as a numeric code.. Without decoding all of these the data was unreadable and hard to visualize.

What I tried: I prompted Claude to read the script and recode the data so it will be easier to translate into the visualizations.

The solution: I asked Claude to write a script that read the value label dictionaries embedded in the Stata file and it automatically mapped the numeric codes to readable labels across all key variables.

How AI helped: I found that recoding the variables would have taken a substantial amount of time. Claude did it within minutes and provided accurate data.

Challenge 4: One of my data sources was in the wrong format

Problem: When I downloaded the 2023 TEDS dataset it was in a format that was not compatible with Python. The initial download was in SAS CPORT format (.stc) which Python cannot read without SAS software installed. The 2021 and 2022 TEDS files downloaded correctly in Stata format without any issues.

What I tried: I prompted Claude to attempt to read the SAS file using pyreadstat's SAS reader and pandas read_sas but both failed with different error messages.

The solution: I went back to SAMHSA's website and downloaded the Stata version of the 2023 TEDS file instead. That version worked perfectly and matched the same format as the 2021 and 2022 files.

How AI helped: Claude identified exactly what each error message meant, explained why the SAS CPORT file was unreadable without SAS software, and pointed me to the right version to download from the SAMHSA website.

Section 8: AI Usage Throughout Project

Which AI Tools I Used:

The only AI tool I used throughout this project was Claude by Anthropic. I used it for data exploration, data cleaning, dashboard development, and debugging.

Where AI Was Most Helpful

The single most helpful thing Claude did was write and debug all the HTML, CSS, and JavaScript for the dashboard. I am not super familiar with the technical aspects of cleaning data and writing code, but Claude was very helpful making that happen. For example, the original NSDUH dataset has 2,638 variables all stored as numeric codes. Translating those by hand would have taken a substantial amount of time and a lot of coding experience, but Claude was able to clean the data within seconds. Claude also made it possible to build out the interactive features of my dashboard through simple prompts such as "Add a year filter feature to all the visualizations with animations corresponding to the year range." That kind of back and forth made the entire development process significantly faster and more manageable than it would have been on my own.

Specific examples of prompts:

Prompt 1: "scan all 2,638 column names in this dataset for anything related to nicotine or tobacco"

Prompt 2: "add hover tooltips to Chapter 6 that show not just the completion rate but also a plain English explanation of what each treatment type actually involves"

Prompt 3: "the 2023 TEDS file is throwing an error that says Header record indicates a CPORT file which is not readable. What does that mean and how do I fix it?"

Where AI Struggled:

One area where Claude struggled was the initial dataset read. When I instructed Claude to make my initial visualizations I noticed some of the numbers seemed lower than expected. I prompted Claude to explain why the data seemed incorrect and it turns out it used the sample dataset rows to create the charts rather than the full dataset. I prompted it to use the full dataset to create the most accurate visualizations and it did, but I did not expect it to not provide the full information from the start.

A second area where Claude struggled is when I prompted it to make a specific change to the dashboard such as "change the background color from its original color to navy blue" and it would also make changes that I did not request, such as slightly changing the title of a page.

Prompting Strategies:

The biggest lesson I learned about working with AI like Claude is that being specific with the changes you want and the changes you do not want is very important. Prior to this class I assumed that AI models would know exactly what I wanted to create and there would be little to no issues, but I ended up having to make a lot of iterations to get the dashboard to look the way I had in mind. I used prompting techniques such as pre-identifying my targets and being specific about what I wanted to show. For example when building my initial third visualization I prompted Claude with "build me a side by side animated bar chart comparing male and female respondents with specifics such as including alcohol use, marijuana use, depression rate, and mental health treatment rate. Additionally, include a caveat explaining that fentanyl, cocaine, and heroine are not included due to extremely low reported rates." Later in the process of creating my dashboard I asked Claude why nicotine and cigarettes were not included and prompted it to rescan the dataset.

Section 9: User Testing & Feedback

Testing Approach:

My user testing was informal. I shared my dashboard with a friend and had her look at it from her phone on Safari to make sure the file was compatible with different devices and browsers. Without providing her with any prior knowledge about the purpose of the dashboard, I had her review it and tell me what she thought the message was. She understood what I was trying to convey, which was a good sign that the story was coming through clearly.

She liked the dashboard overall, but was confused by the different treatment type labels in Chapter 6 such as "Detox Hospital" and "Detox Residential". I realized that for my general public audience this would likely be a common issue since most people are not familiar with the different types of substance abuse treatment. I prompted Claude to add hover tooltips to every bar in that chapter that provide an explanation of what each treatment type entails, so users never have to leave the dashboard to understand what they are looking at.

Section 10: Reflection & Future Directions

Personal Learning:

I learned that there is a really extensive process for cleaning data before you even get to the visualization step, and I cannot imagine how much work that would require without the help of AI. It gave me a much deeper appreciation for what goes into data analysis at a professional level, especially before AI was introduced. I am really proud of the dashboard I created and I plan on presenting it to my dad when I get home for the summer. It was meaningful to be able to combine a personal interest with a class project and be able to discuss and share it with a friend. For future additions, I would like to include more information to make the findings even more accurate and complete. For example, having data on substance use rates from incarcerated and homeless communities would have been really valuable since these are groups at a higher risk for substance abuse. Their absence from the dataset is not just a data limitation but it also reflects a bigger issue of underrepresented groups not receiving help and that starts with a lack of data being collected on them in the first place. I also would have liked to explore how geography plays into substance use more and answer questions like which substances are most prevalent in different states or whether drug use is more common in rural versus urban areas. Adding the

physical health effects of long term substance use such as liver failure and brain damage would have also strengthened the case for why this issue is so important. This approach could also be applied to many other systematic issues found in the American system such as disproportionate literacy rates across demographics or incarceration rates broken down by demographic and age group.

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AI Disclosure Statement

I pledge on my honor that I have not given or received any unauthorized assistance on this assignment/examination. I further pledge that I have not copied any material from a book, article, the Internet or any other source except where I have expressly cited the source. I have documented all AI tool usage including prompts and tool selection rationale.

Signature: _____ Tory Seaton _____ Date: _____ May 9th 2026 _____